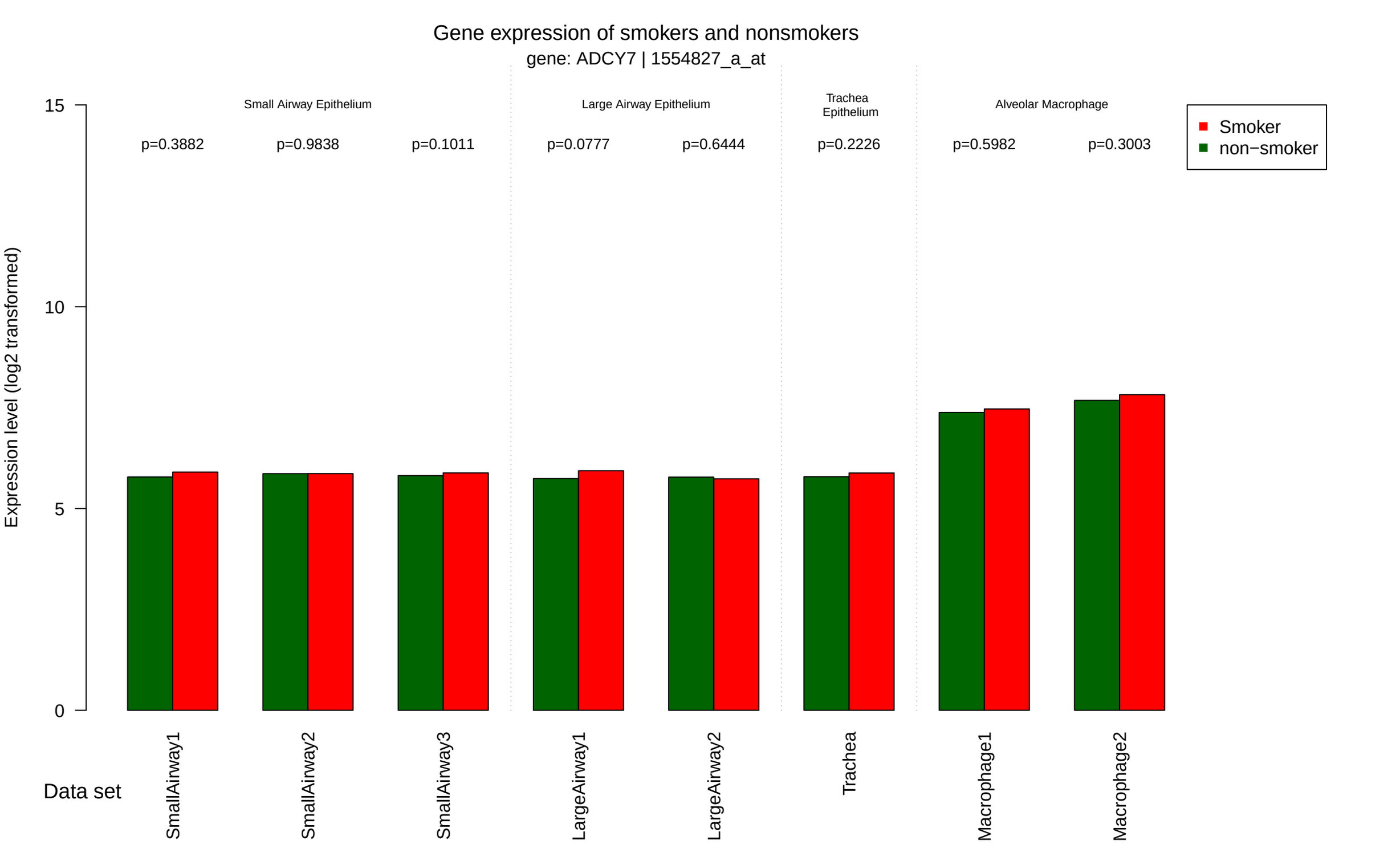
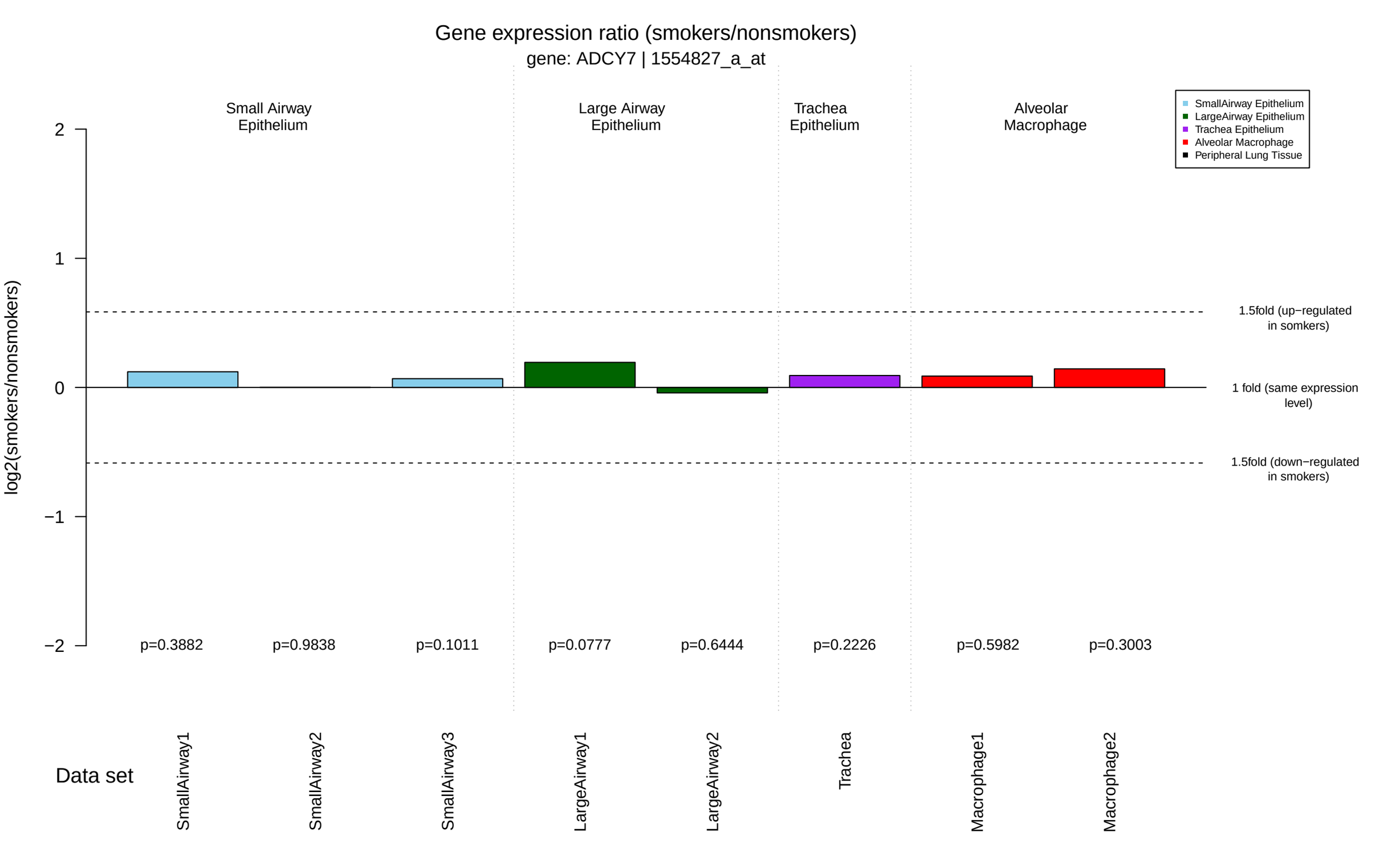
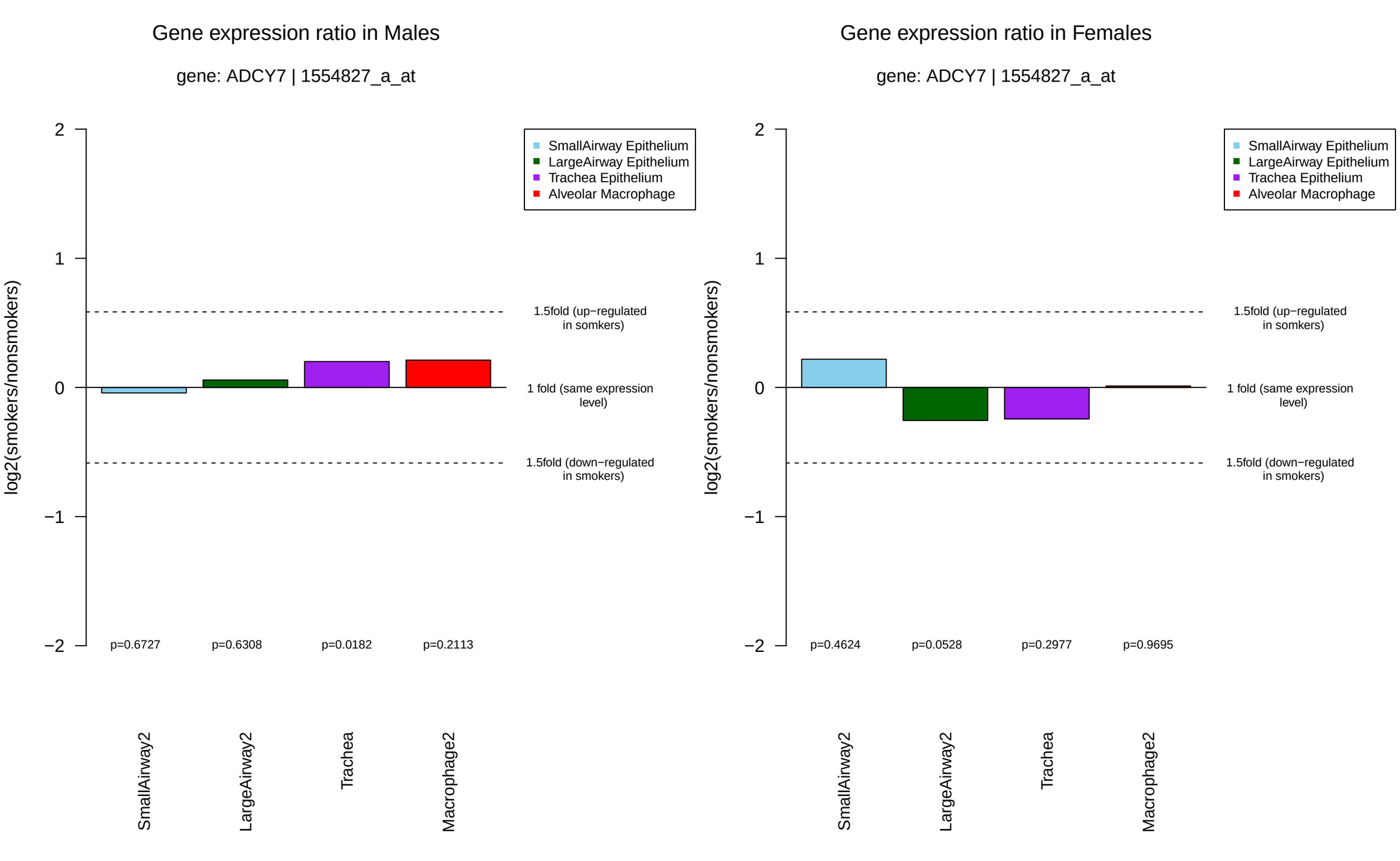
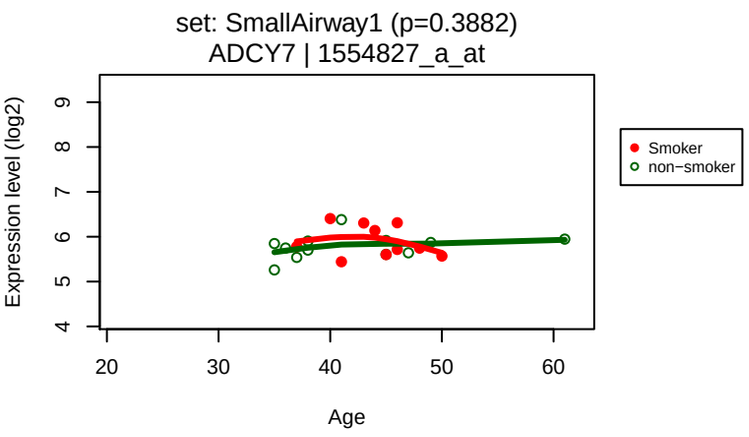
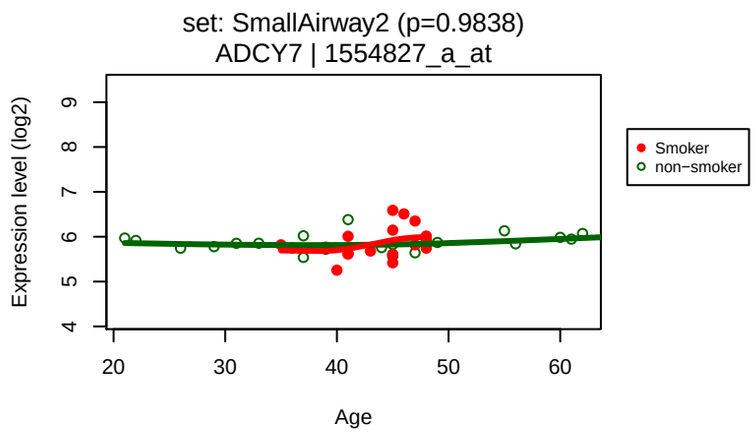
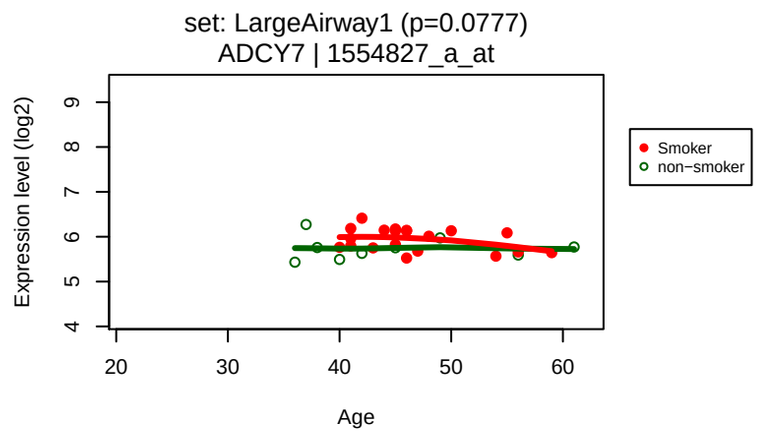

Gene expression of smokers and nonsmokers
gene: ADCY7 | 1554827_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3882
p=0.9838
p=0.1011
p=0.0777
p=0.6444
p=0.2226
p=0.5982
p=0.3003
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

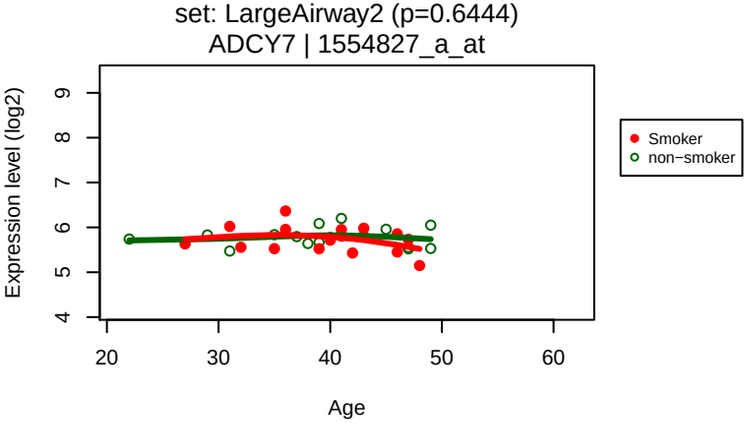

Gene expression ratio (smokers/nonsmokers)
gene: ADCY7 | 1554827_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3882
p=0.9838
p=0.1011
p=0.0777
p=0.6444
p=0.2226
p=0.5982
p=0.3003
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

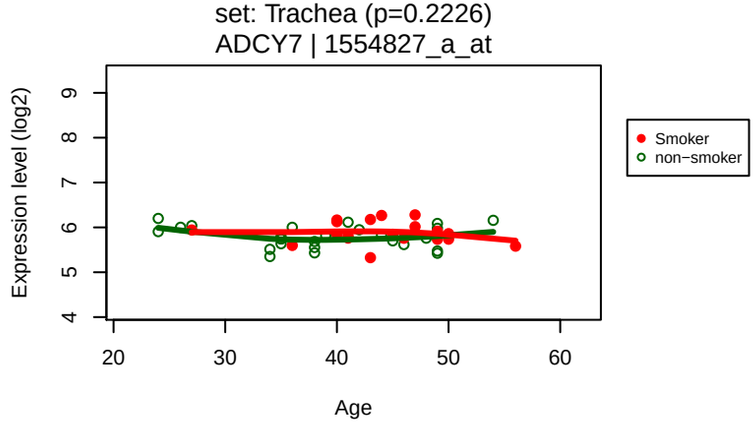

Gene expression ratio in Males
gene: ADCY7 | 1554827_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6727
p=0.6308
p=0.0182
p=0.2113
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY7 | 1554827_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4624
p=0.0528
p=0.2977
p=0.9695
SmallAirway2
LargeAirway2
Trachea
Macrophage2

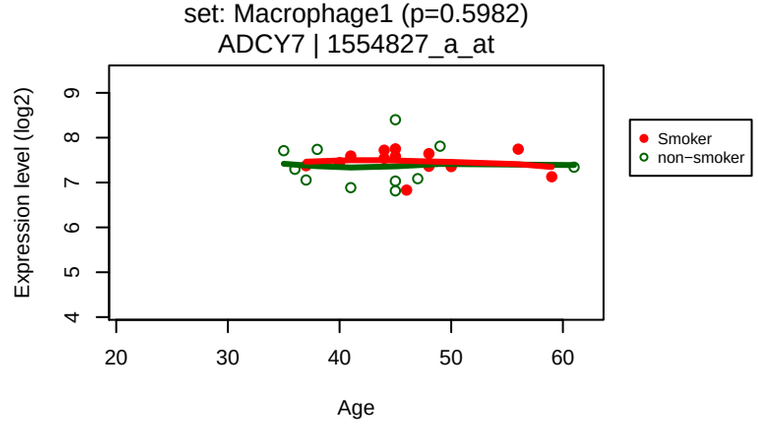

set: SmallAirway1 (p=0.3882)
ADCY7 | 1554827_a_at
Expression level (log2)
Age
Smoker
non−smoker

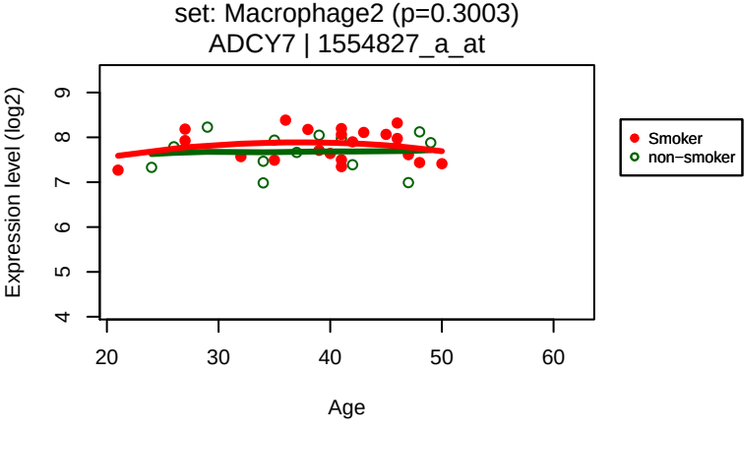

set: SmallAirway2 (p=0.9838)
ADCY7 | 1554827_a_at
Expression level (log2)
Age
Smoker
non−smoker

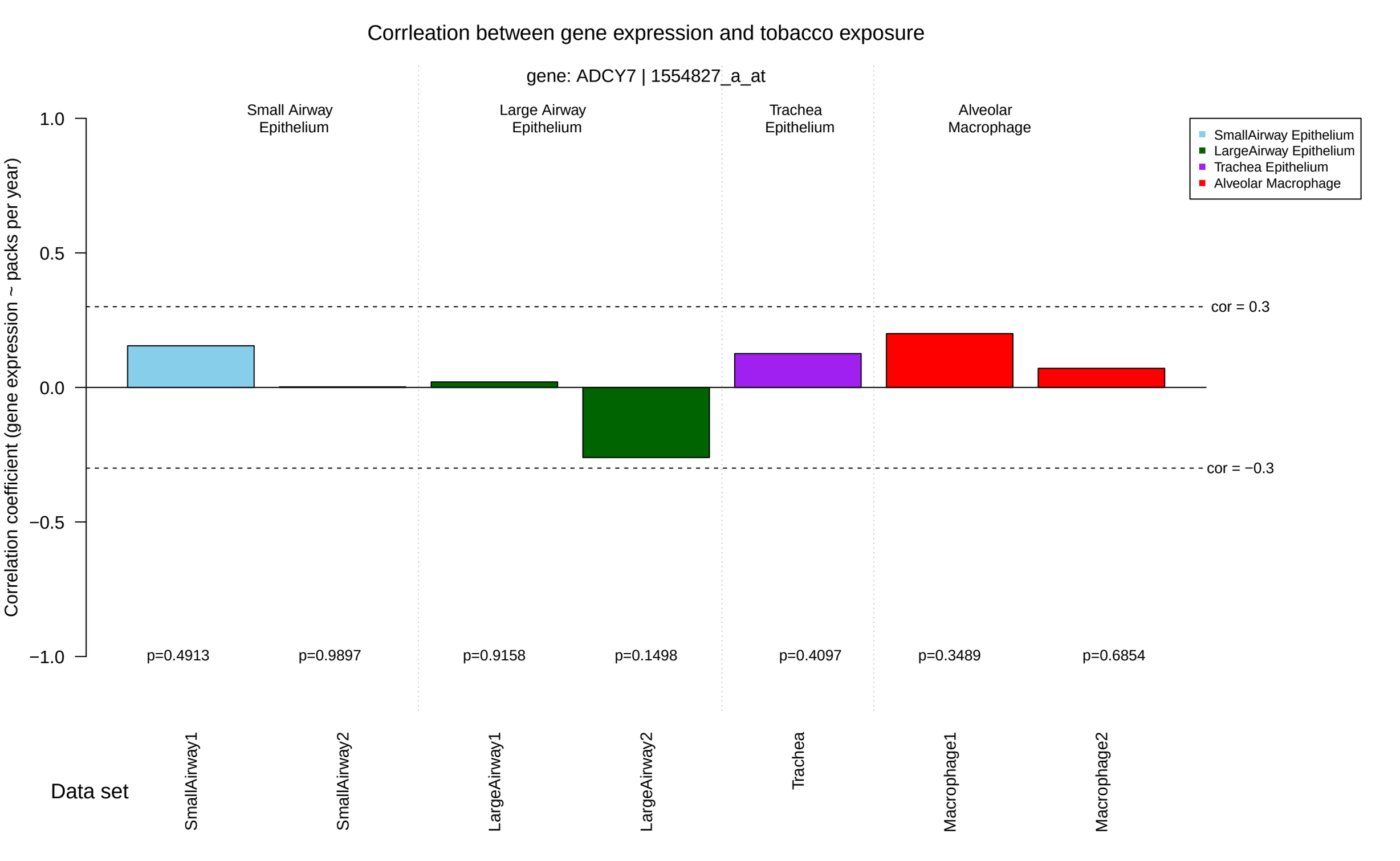

set: LargeAirway1 (p=0.0777)
ADCY7 | 1554827_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6444)
ADCY7 | 1554827_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2226)
ADCY7 | 1554827_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5982)
ADCY7 | 1554827_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3003)
ADCY7 | 1554827_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY7 | 1554827_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4913
p=0.9897
p=0.9158
p=0.1498
p=0.4097
p=0.3489
p=0.6854
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2